

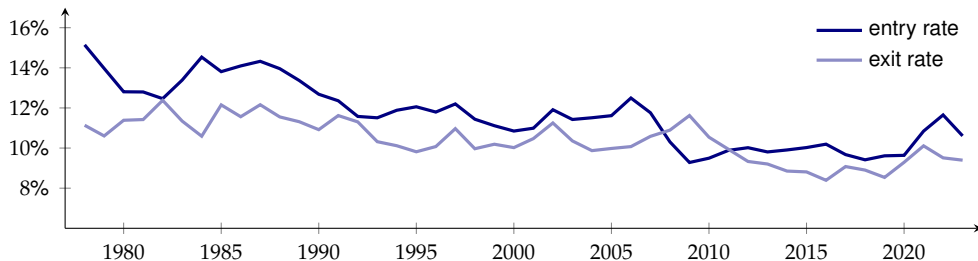
Entry, Exit, and Firm Dynamics in Long Run Equilibrium

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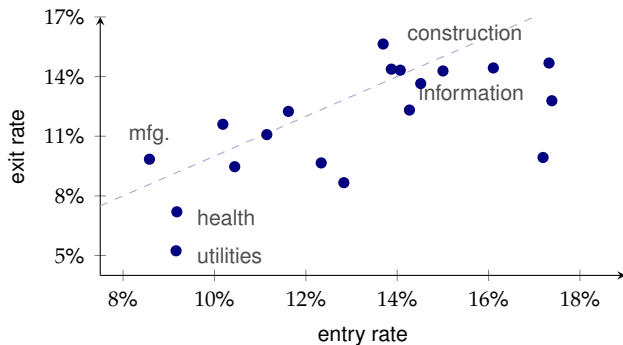
Entry and exit are large, and they never stop



- Roughly **one establishment in ten** enters each year, and about as many leave. Neither rate ever goes near zero.
- The two **move together**, and exit overtakes entry only in recessions (1982, 2008–09).
- Firm-specific shocks, not industry-wide ones, dominate firm-level dynamics.

U.S. Census Bureau, Business Dynamics Statistics, 1978–2023, establishment entry and exit rates

Turnover is an industry characteristic



Each point is a NAICS sector.

- Entry and exit rates are **strongly correlated across industries**: the correlation is 0.65.
- There are **high-turnover** industries (construction, accommodation, information) and **low-turnover** ones (utilities, health, manufacturing).
- A theory of turnover must explain a persistent *level* that differs across industries, not a transition that dies away.

Business Dynamics Statistics, 19 NAICS sectors, 2011

The gap in long-run industry theory

Marshall; Lucas (1978)	Jovanovic (1982)	Hopenhayn (1992)
long run = a static population of firms no entry or exit, no heterogeneity in growth	selection through passive learning entry and exit vanish in the limit	selection through a persistent shock entry and exit continue forever

The equilibrium question

How can a competitive industry be in a *steady state* while firms keep entering, growing, shrinking, and leaving?

The answer has to make the *distribution* of firms the equilibrium object, and keep the flow through it constant.

What the paper does

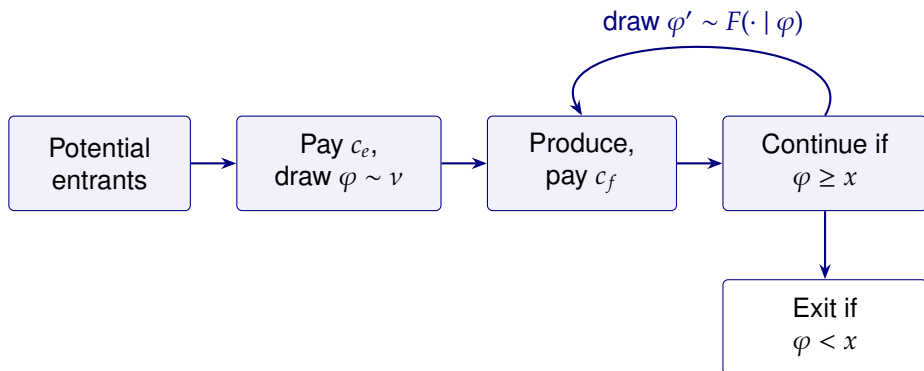
Develops a **dynamic stochastic model of a competitive industry** in which entry, exit, and each firm's output and employment are determined *endogenously*, and studies its **steady state**.

Three contributions

- **A concept:** *stationary equilibrium*, extending long-run industry equilibrium to entry, exit, and heterogeneity in size and growth. Existence and uniqueness.
- **Conditions** under which that equilibrium has *positive* entry and exit. It need not: Jovanovic's model is the counterexample.
- **Comparative statics** of technology and market conditions on the size distribution, turnover, profits, and firm values.

The engine throughout is **selection**: an endogenous exit cutoff truncates the shock distribution every period.

The mechanism in one picture



A constant inflow of entrants, a persistent shock that reshuffles them, and a cutoff that removes the weakest. The three together hold the cross-sectional distribution fixed while no individual firm stands still.

A stationary industry is not a stationary firm.

Environment

A continuum of firms produces a **homogeneous** good. Firms are **competitive**: they take output and input prices as given.

- Aggregate demand is the inverse demand function $D(Q)$; the input price is $W(N)$, with N total industry demand for the input.
- One input (“labor”); the results extend to several inputs under standard homotheticity assumptions.

Assumption A.1

- (a) D is continuous, strictly decreasing, with $\lim_{x \rightarrow \infty} D(x) = 0$.
- (b) W is continuous, nondecreasing, and strictly bounded above zero.

Today we assume more. We adopt the paper’s **Condition U.1**: the industry is a price taker in its input market, facing perfectly elastic supply at $\underline{w}$, so $w = \underline{w}$ throughout. Hopenhayn lets W slope up; holding it flat buys uniqueness and a recursive structure, at the cost of general-equilibrium price effects.

Technology and the idiosyncratic shock

A firm's output is

$$q = f(\varphi, n), \quad \varphi \in S = [0, 1],$$

where φ is a **productivity shock** following a Markov process, *independent across firms*, with conditional distribution $F(\varphi' \mid \varphi)$.

- Every incumbent pays a **fixed cost** c_f each period, so at prices (p, w) profit is

$$\pi(\varphi, p, w) = \max_{n \geq 0} \left[p f(\varphi, n) - w n \right] - c_f,$$

- with $n(\varphi, p, w)$ the input demand and $q(\varphi, p, w) = f(\varphi, n(\varphi, p, w))$ the output supply.
- $\pi(\varphi, 0, w) = -c_f$: **a positive fixed cost is necessary for exit**. It is equivalent to a fixed outside opportunity cost used by the firm.

Assumption A.2

(a) q and n are single valued, continuous, strictly increasing in φ ; (b) π is continuous and strictly increasing in φ ; (c) $\lim_{Q \rightarrow 0} \pi(0, D(Q), w) > 0$.

Assumptions on the shock process

Assumption A.3

(a) F is continuous in φ and φ' ; (b) F is *strictly decreasing* in φ .

A.3(b) is first-order stochastic dominance: higher today makes higher tomorrow more likely. With A.2 it makes expected discounted profits increasing in φ , which delivers a **cutoff** exit rule.

Assumption A.4 (Recurrence)

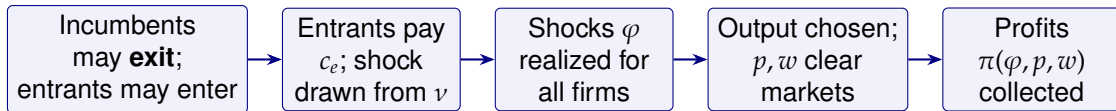
For any $\varepsilon > 0$ there is an integer n with $F^n(\varepsilon \mid \varphi) > 0$, where $F^n(\cdot \mid \varphi)$ is the distribution of φ_{t+n} given $\varphi_t = \varphi$.

Why recurrence? It makes a firm's life span *almost surely finite*, so every cohort eventually empties out.

Assumption A.5

ν , the initial distribution of entrants' shocks, has a continuous distribution function.

Timing within a period



- A firm that exits secures a present value **normalized to zero**: no scrap value, no resale.
- The entry cost $c_e \geq 0$ is paid *before* the shock is revealed. Entry is a draw from ν , not a choice of φ .
- **Crucially**: the exit decision is made *prior to* observing next period's shock. The cutoff is on the *current* state φ .

The state of the industry

Let μ_t be a **measure over firms' shocks**: for a Borel set $A \subset S$, $\mu_t(A)$ is the mass of firms with shocks in A , and $\mu_t(S)$ is the total mass. Aggregating supply and input demand,

$$Q^s(\mu, p, w) = \int q(\varphi, p, w) \mu(d\varphi), \quad N^d(\mu, p, w) = \int n(\varphi, p, w) \mu(d\varphi).$$

No aggregate uncertainty

Shocks are independent across firms and there are no industry-wide shocks, so the cross-sectional frequency distribution coincides with the probability distribution. Aggregate output, employment, prices, and the distribution of φ follow **deterministic paths**; only a firm's *own* φ_t is stochastic.

An industry is the list $\{D(\cdot), W(\cdot), c_f, c_e, f, S, F, \nu, \mu_0\}$. *Warning*: the paper writes $M_t = \mu_t(S)$ here, but later uses M_t for the mass of **entrants**. We reserve M for entrants and write $\mu(S)$ for the total mass.

The incumbent's problem

Firms have **perfect foresight** over the deterministic price sequence $z = \{p_t, w_t\}$ and discount at $\beta \in (0, 1)$. The incumbent's problem is recursive:

$$v_t(\varphi, z) = \pi(\varphi, p_t, w_t) + \beta \max\left\{0, \int v_{t+1}(\varphi', z) F(d\varphi' \mid \varphi)\right\}.$$

- v_t is the value of a type- φ firm *after* its new shock is realized.
- The $\max\{0, \cdot\}$ is the **exit option**: continuation is worth zero if abandoned.
- The expectation sits *inside* the max, because exit is decided before φ' is seen.

Proposition 1 (Properties of v_t)

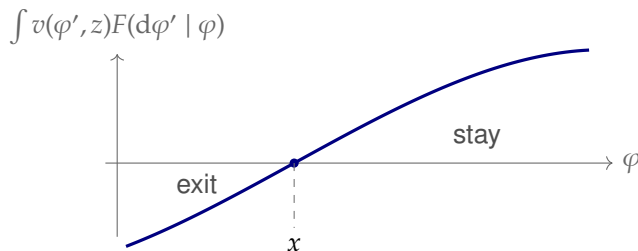
For a bounded price sequence, v_t is (i) strictly increasing and continuous in φ ; and (ii) the continuation value $\int v_{t+1}(\varphi', z) F(d\varphi' \mid \varphi)$ is strictly increasing in φ .

Exit is a reservation-productivity rule

Because the continuation value is strictly increasing in φ , the exit set is an interval: firms exit whenever the state falls below a **reservation level**

$$x_t = \inf\{\varphi \in S : \int v_{t+1}(\varphi', z)F(d\varphi' | \varphi) \geq 0\},$$

so an interior cutoff satisfies $\int v_{t+1}(\varphi', z)F(d\varphi' | x_t) = 0.$



- Selection is **one-dimensional**: a single number summarizes the entire exit decision.
- By A.4 the cutoff is hit almost surely, so expected lifetimes are finite.

Prop. 1. Figure schematic.

Entry and the law of motion

Entry. Potential entrants are identical and pay c_e before drawing $\varphi \sim \nu$. The value of entering is $v_t^e(z) = \int v_t(\varphi, z) \nu(d\varphi)$, and **free entry** is a complementary-slackness condition,

$$v_t^e(z) \leq c_e, \quad \text{with equality if } M_t > 0.$$

Law of motion. Survivors above the cutoff redraw through F , and the entering cohort is added:

$$\mu_{t+1}(d\varphi') = \int_{\varphi \geq x_t} F(d\varphi' | \varphi) \mu_t(d\varphi) + M_{t+1} \nu(d\varphi')$$

Writing $\widehat{P}_t(\varphi, A) = \int_A F(ds | \varphi)$ for $\varphi \geq x_t$ and 0 otherwise,

$$\mu_{t+1} = \widehat{P}_t \mu_t + M_{t+1} \nu.$$

Industry equilibrium

A sequence $\{p_t, w_t, Q_t, N_t, M_t, x_t, \mu_t\}$ with (i) market clearing, (ii) x_t optimal given prices, (iii) free entry, and (iv) the law of motion from μ_0 .

The equilibrium path exists and is unique

Theorem 1

Given any initial distribution there exists a **unique** equilibrium.

Method. The equilibrium solves a *planning problem* in (Q_t, N_t) : maximize the discounted sum of consumer plus producer surplus, net of entry and fixed costs,

$$\max \sum_t \beta^t [R(Q_t) - C(N_t) - c_e M_t - c_f \mu_t(S)], \quad R' = D, \quad C' = W.$$

Existence follows from compactness and continuity; the objective is **strictly concave** in (Q, N) because D is strictly decreasing, so the solution, and hence the price path, is unique.

- Unique prices $\Rightarrow$ exit rules coincide, and recursively so do the entry masses.
- The sequence problem defines a dynamical system $\mu_{t+1} = H(\mu_t)$; a stationary equilibrium is a **steady state** of H .
- Equilibria maximize net discounted surplus: no distortion for policy to correct.

Stationary equilibrium: the object

A **stationary equilibrium** is a vector $(p^*, w^*, Q^*, N^*, x^*, M^*, \mu^*)$ such that the constant sequence it generates is an equilibrium starting from $\mu_0 = \mu^*$.

Given any μ with $\mu(S) > 0$, market clearing defines prices $p^e(\mu)$, and $\tilde{\pi}(\varphi, \mu)$, $v(\varphi, \mu)$ denote current profit and firm value at those prices, with

$$v(\varphi, \mu) = \tilde{\pi}(\varphi, \mu) + \max\left\{0, \beta \int v(\varphi', \mu) F(d\varphi' \mid \varphi)\right\}.$$

Propositions 2–3

$\tilde{\pi}$ is jointly continuous, strictly increasing in φ , and **decreasing in μ** . There is a unique continuous v solving it; it is strictly increasing in φ , decreasing in μ , and its continuation value is strictly increasing in φ .

Decreasing in μ is the force behind existence: more firms means a lower price, hence a lower value of being in the industry.

The three stationary conditions

A stationary equilibrium with positive entry is a triple (x, M, μ) solving:

Exit: the marginal firm is indifferent

$$\int v(\varphi', \mu) F(d\varphi' \mid x) = 0$$

Free entry: expected value equals the entry cost

$$\int v(\varphi, \mu) \nu(d\varphi) = c_e$$

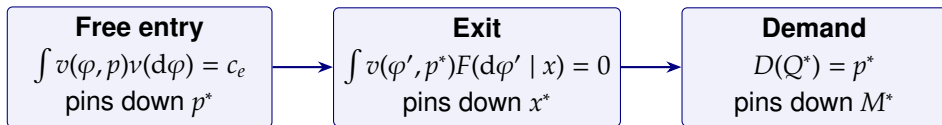
Consistency: μ is invariant under (x, M)

$$\mu = \widehat{P}_x \mu + M\nu$$

Three equations in three unknowns. Prices do not appear: they are substituted out through $p^e(\mu)$.

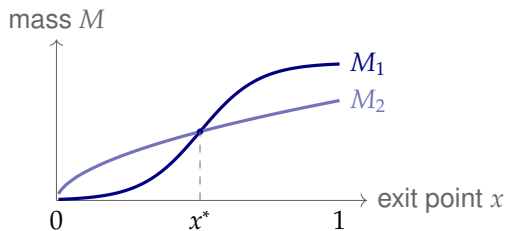
Why the model is tractable: block recursivity

Because the input price is fixed by U.1, $v(\varphi, \mu)$ depends on μ *only through the output price*, and the system unwinds in one direction.



- **The price is a technological object.** It is set by the entry condition alone, through c_e , c_f , F , v , β , and not by demand. The long-run supply curve is horizontal.
- Demand then chooses only the *scale* of the industry, so every cross-sectional and life-cycle property is independent of demand.
- With an upward-sloping input supply both prices would depend on μ and this separation would break.

Existence: two curves in (x, M) space



Let $m(x, M)$ be the measure invariant under exit rule x and entry mass M . For $x \in (0, 1]$ define two entry masses.

$M_1(x)$ makes x **optimal**:

$$\int v[\varphi, m(x, M_1(x))] F(d\varphi | x) = 0$$

$M_2(x)$ makes **entry break even**:

$$\int v(\varphi, m(x, M_2(x))) \nu(d\varphi) = c_e$$

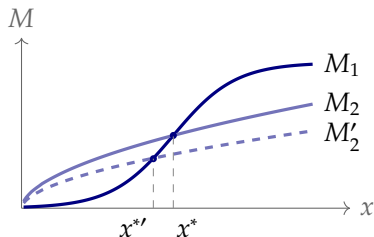
M_1 is continuous and strictly increasing, M_2 continuous and nondecreasing, so positive entry exists **iff** the curves cross. Since v peaks at $\varphi = 1$, $M_1(1) > M_2(1)$: either they cross, or only $M = 0$ equilibria exist.

Theorems 2–4

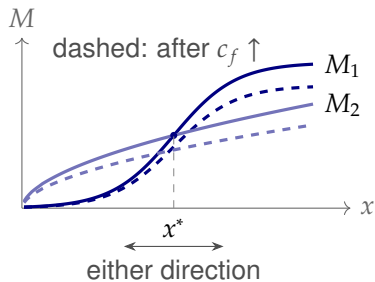
A stationary equilibrium exists; one with **positive entry** exists *iff* $c_e < c^*$; under U.1 the crossing is **unique**.

Comparative statics: entry cost and fixed cost

Higher entry cost c_e



Higher fixed cost c_f



- $c_e \uparrow$: only M_2 falls, so the crossing slides **left and down**. Cutoff, entry, and turnover all fall; weaker firms survive and incumbents are protected (Orr 1974).
- $c_f \uparrow$: **both** curves fall, and whichever falls more sets the direction. Separable profits raise the cutoff; the paper's example lowers it.

The invariant distribution has a closed form

By A.4 the inverse of $(I - \widehat{P}_x)$ exists, so the invariant measure for exit rule x and entry mass M is

$$\mu = M(I - \widehat{P}_x)^{-1}v = M \sum_{t=0}^{\infty} \widehat{P}_x^t v.$$

With λ_t the probability of surviving t periods after entry and $\bar{\mu}_t$ the shock distribution of that cohort,

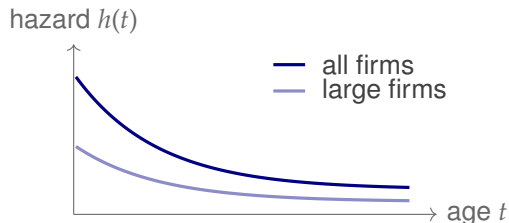
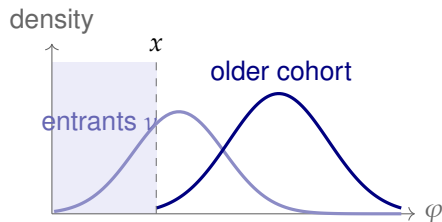
$$\sum_{t=0}^{\infty} \widehat{P}_x^t v(A) = \sum_{t=0}^{\infty} \lambda_t \bar{\mu}_t(A).$$

The industry is a superposition of cohorts

Setting $A = S$ gives $\mu^*(S) = M^* \sum_{t \geq 0} \lambda_t$: the mass of firms is the entry flow times the expected lifetime, so the **turnover rate is the reciprocal of expected firm lifetime**,

$$\rho \equiv M^* / \mu^*(S) = \left(\sum_{t \geq 0} \lambda_t \right)^{-1}.$$

Life cycle: selection sorts firms by age and size



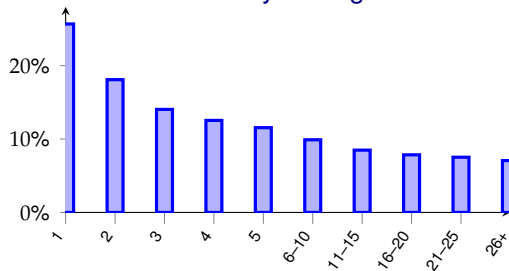
The hazard is $h(t) = 1 - \lambda_{t+1}/\lambda_t$: the chance of exiting at age t having reached it.

- Every cohort dominates the entrant distribution, so any increasing function — value, profits, survival — is at its minimum for the newest entrants.
- **Hazard rates are highest for the youngest cohort**, exactly the survival pattern reported by Dunne, Roberts, and Samuelson.
- **Larger firms are longer lived** and tend to remain larger before they exit, so hazard rates fall with size as well as with age.

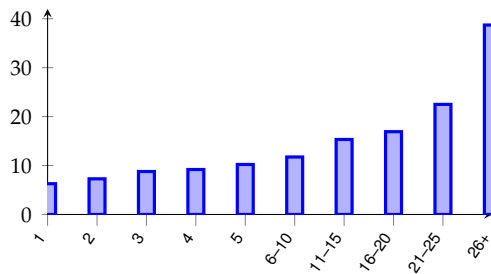
Figures schematic.

Selection in the data

Exit rate by firm age



Employees per firm by age



- The exit rate **falls monotonically with age**, from 26% in the first year to 7% after twenty-five.
- Surviving cohorts are **progressively larger**: six employees at age one, thirty-nine at age twenty-six and above.

BDS 2023 by firm age; exits over mean start/end establishments. Age 0 omitted: degenerate.

Profits and the value of firms

Incumbents hold an **option**: they may stay without paying a new entry cost. So the marginal firm can run losses, $\tilde{\pi}(x^*, \mu^*) < 0$, yet average profits are bounded below.

Proposition 7

$$\bar{\pi}(\mu^*) \equiv \int \tilde{\pi}(\varphi, \mu^*) \mu^*(d\varphi) \geq \rho c_e.$$

Proposition 8

$$\bar{v}(\mu^*) = \frac{\bar{\pi}(\mu^*) - \beta \rho c_e}{1 - \beta} \geq \rho c_e.$$

Why this matters. Without turnover $\bar{v} = \bar{\pi}/(1 - \beta)$; turnover opens a **gap**, because the profit flow must also cover entry costs. The gap is an *indirect measure of sunk entry costs*, vanishing as $\rho \rightarrow 0$.

As printed on p. 1145 the bracket is misplaced; the proof gives this form

Takeaways

- **A tractable steady state for an industry with turnover.** Entry, exit, and the size distribution are endogenous, the equilibrium exists and is unique, yet the object to solve for is just (x^*, M^*, μ^*) .
- **Free entry prices the industry.** The entry condition pins down p^* from technology and costs; demand sets scale.
- **Selection is the mechanism.** One reservation productivity delivers hazard rates declining in age and in size, and cohorts that improve with age — as theorems, not calibration.
- **Turnover is the reciprocal of expected lifetime,** $\rho = (\sum_t \lambda_t)^{-1}$, falling with the entry cost.
- **Sunk costs are identified by a value–profit wedge** (Proposition 8), a measurable implication.

“Entry and exit are part of the limiting behavior of the industry and not only part of the adjustment to a steady state.”

Legacy

- **The workhorse of firm dynamics.** “The Hopenhayn model” is now the standard block for entry and exit in macro. Hopenhayn–Rogerson (1993) adds labor adjustment costs and general equilibrium, and prices firing taxes.
- **Misallocation.** The stationary distribution over productivities is exactly the object distorted in Restuccia–Rogerson (2008) and Hsieh–Klenow (2009); entry and exit are where policy distortions accumulate.
- **Trade.** Melitz (2003) is the open-economy cousin: the same free-entry-plus-cutoff logic, but with a *static* productivity draw, so there is no growth, no age effect, and exit is driven by fixed costs rather than by a Markov process.
- **What the transition adds.** $F(\varphi' \mid \varphi)$ generates a genuine *life cycle* — survival curves, cohort effects, growth–size relations — and makes expected lifetime an equilibrium object.

Fit, and what the paper leaves out

- **Partial equilibrium.** $D(\cdot)$ and $W(\cdot)$ are primitives: no household, no other sectors.
- **No policy question.** The equilibrium already maximizes net discounted surplus.
- **Only one dynamic decision.** Exit. No investment, no innovation, no learning.
- **Score against the data.** With a suitable F : regression to the mean in size, and higher persistence for larger firms (Evans 1987; Dunne–Roberts–Samuelson; Leonard 1987).
- **No aggregate shocks**, so no business cycle in entry and exit.